

DCCN LAB ASSIGNMENT - 2

Siddharth Narayan Mishra

123CS0189

Q1: Answer the following questions for captured file http.pcap (HTTP Protocol)

1. List 3 different protocols that appear in the protocol column in the unfiltered packet-listing window.

TCP, HTTP and DNS are the 3 different protocols that appear in the protocol column in the unfiltered packet-listing window.

2. How long did it take from when the HTTP GET message was sent until the HTTP OK reply was received?

I choose packet number 217 for GET request and 219 for the OK reply.

Timestamp for packet 217 = 12:49:01.271363

Timestamp for packet 219 = 12:49:01.078947

Time taken = 12:49:01.271363 - 12:49:01.078947 = 0.192416 seconds

It is also mentioned in "Time since request" field in HTTP section of the OK packet.

3. What is the Internet address of iitd.ac.in? What is the Internet address of your computer?

Internet address of iitd.ac.in = 103.27.9.167

Internet address of my computer = 127.0.1.1

4. Print the two HTTP messages (GET and OK) referred to in question 2 above. Done. PDF files attached below.

5. Find the packet number that includes HTTP GET message for a file IITD-IRD-122-2017.pdf. Also find the length of the file in bytes and time when file is downloaded successfully.

Packet Number = 478

Length of File = 18897 bytes

Time of Download = Jul 29, 2017 12:49:13.700007000 IST

Q2: Write a C/C++/Java/Python code to read the data in csv file and print

- a. source IP addresses and destination IP addresses
- b. source port numbers and destination port numbers
- c. http request and response messages.

CODE: (python)

```
import pandas as pd
import re

df = pd.read_csv("http.csv")

port_pattern = re.compile(r'(\d+)\s*>\s*(\d+)')

http_keywords = ["GET", "POST", "HEAD", "PUT", "DELETE",
"HTTP/1.1"]

with open("output.txt", "w") as f:
    header = (
        f"{'Source IP':<20}"
        f"{'Destination IP':<20}"
        f"{'Source Port':<15}"
        f"{'Destination Port':<20}"
        f"{'HTTP Request / Response':<40}\n"
    )

    f.write(header)
    f.write("=" * 115 + "\n")

    for _, row in df.iterrows():
        src_ip = str(row['Source'])
        dst_ip = str(row['Destination'])
```

```
src_port = "-"
dst_port = "-"
http_info = "-"
match = port_pattern.search(str(row['Info']))

if match:
    src_port, dst_port = match.groups()

info = str(row['Info'])
protocol = str(row['Protocol'])
if protocol == "HTTP" or any(k in info for k in
http_keywords):
    http_info = info

f.write(
    f"{src_ip:<20}"
    f"{dst_ip:<20}"
    f"{src_port:<15}"
    f"{dst_port:<20}"
    f"{http_info:<40}\n"
)

print("output saved to output.txt")
```

OUTPUT SCREENSHOT:

(saved in output.txt, showing a part of the output)

```
output.txt (~/.work/acads/sem6/dccn/lab2)
File Edit View Search Tools Documents Help
output.txt x
Source IP      Destination IP    Source Port      Destination Port  HTTP Request / Response
-----
XiaomiCommun_9e:9c:c3Broadcast - - -
HonHaiPrecis_8c:90:55XiaomiCommun_9e:9c:c3- - -
192.168.43.153 103.27.9.20 34573 80 -
192.168.43.153 192.168.43.1 - -
192.168.43.1 192.168.43.153 - -
192.168.43.153 103.27.9.20 34574 80 -
103.27.9.20 192.168.43.153 80 34574 -
192.168.43.153 103.27.9.20 34574 80 -
192.168.43.153 103.27.9.20 - GET / HTTP/1.1
103.27.9.20 192.168.43.153 80 34573 -
192.168.43.153 103.27.9.20 34573 80 -
103.27.9.20 192.168.43.153 80 34574 -
103.27.9.20 192.168.43.153 80 34574 -
192.168.43.153 103.27.9.20 34574 80 -
103.27.9.20 192.168.43.153 80 34574 -
192.168.43.153 103.27.9.20 34574 80 -
103.27.9.20 192.168.43.153 80 34574 -
192.168.43.153 103.27.9.20 34574 80 -
192.168.43.153 103.27.9.20 - GET /sites/default/files/ctools/css/2d4fe1c1412881a73110e50a115127de.css?Y HTTP/1.1
103.27.9.20 192.168.43.153 34575 80 -
103.27.9.20 192.168.43.153 80 34574 -
192.168.43.153 103.27.9.20 34574 80 -
192.168.43.153 103.27.9.20 34576 80 -
192.168.43.153 103.27.9.20 34577 80 -
192.168.43.153 103.27.9.20 34578 80 -
103.27.9.20 192.168.43.153 80 34574 -
192.168.43.153 103.27.9.20 34574 80 -
103.27.9.20 192.168.43.153 80 34574 -
192.168.43.153 103.27.9.20 - HTTP/1.1 200 OK (text/html)
192.168.43.153 103.27.9.20 34574 80 -
192.168.43.153 103.27.9.20 - GET /modules/node/node.css?Y HTTP/1.1
103.27.9.20 192.168.43.153 80 34576 -
192.168.43.153 103.27.9.20 34576 80 -
192.168.43.153 103.27.9.20 - GET /modules/system/defaults.css?Y HTTP/1.1
103.27.9.20 192.168.43.153 80 34578 -
192.168.43.153 103.27.9.20 34578 80 -
192.168.43.153 103.27.9.20 - GET /modules/system/system-menus.css?Y HTTP/1.1
103.27.9.20 192.168.43.153 80 34575 -
```

No.	Time	Source	Destination	Protocol	Length
217	2017-07-29 12:49:01.078947	192.168.43.153	162.213.33.44	HTTP	195
GET /v0/search HTTP/1.1					
Frame 217: 195 bytes on wire (1560 bits), 195 bytes captured (1560 bits)					
Ethernet II, Src: HonHaiPrecis_8c:90:55 (e0:06:e6:8c:90:55), Dst: XiaomiCommun_9e:9c:c3 (ac:c1:ee:9e:9c:c3)					
Internet Protocol Version 4, Src: 192.168.43.153, Dst: 162.213.33.44					
0100 = Version: 4					
.... 0101 = Header Length: 20 bytes (5)					
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)					
Total Length: 181					
Identification: 0xb7f2 (47090)					
010. = Flags: 0x2, Don't fragment					
...0 0000 0000 0000 = Fragment Offset: 0					
Time to Live: 64					
Protocol: TCP (6)					
Header Checksum: 0xd20d [validation disabled]					
[Header checksum status: Unverified]					
Source Address: 192.168.43.153					
Destination Address: 162.213.33.44					
Transmission Control Protocol, Src Port: 37947, Dst Port: 80, Seq: 1, Ack: 1, Len: 129					
Hypertext Transfer Protocol					

No.	Time	Source	Destination	Protocol	Length
219	2017-07-29 12:49:01.271363	162.213.33.44	192.168.43.153	HTTP/JSON	402
Info					
HTTP/1.1 200 OK , JSON (application/json)					
Frame 219: 402 bytes on wire (3216 bits), 402 bytes captured (3216 bits)					
Ethernet II, Src: XiaomiCommun_9e:9c:c3 (ac:c1:ee:9e:9c:c3), Dst: HonHaiPrecis_8c:90:55 (e0:06:e6:8c:90:55)					
Internet Protocol Version 4, Src: 162.213.33.44, Dst: 192.168.43.153					
0100 = Version: 4					
.... 0101 = Header Length: 20 bytes (5)					
Differentiated Services Field: 0x28 (DSCP: AF11, ECN: Not-ECT)					
Total Length: 388					
Identification: 0xa676 (42614)					
010. = Flags: 0x2, Don't fragment					
...0 0000 0000 0000 = Fragment Offset: 0					
Time to Live: 49					
Protocol: TCP (6)					
Header Checksum: 0xf192 [validation disabled]					
[Header checksum status: Unverified]					
Source Address: 162.213.33.44					
Destination Address: 192.168.43.153					
Transmission Control Protocol, Src Port: 80, Dst Port: 37947, Seq: 1, Ack: 130, Len: 336					
Hypertext Transfer Protocol					
JavaScript Object Notation: application/json					